

Chapter 19: Ports and Connectors —

Notes

Introduction

- **Port:** hardware connector attached to the PC itself.
- **Connector:** end of the cable attached to the peripheral device.
- Standard external ports: serial, parallel, USB/FireWire, video, game device, speaker/mic jacks.

Connectors on the Motherboard

Classified into three groups: 1. **Back panel connectors** – external I/O connectors (focus of chapter) 2. **Midboard connectors** – for internal peripherals 3. **Front panel connectors** – for case switches/LEDs

Onboard Connectors (back panel, common set)

- PS/2 keyboard and mouse connectors
- Two USB ports (stacked)
- One serial port (A)
- Parallel port
- VGA (video)
- MIDI/game port
- Optional audio jacks: Line in, Line out, Microphone in

Midboard Connector Groups

- **Audio/video:** sound, video, CD-ROM support (aux sound line in, telephony, legacy CD-ROM, ATAPI CD-ROM)
- **Peripheral device interfaces:** Primary/Secondary IDE connectors, floppy disk controller
- **Hardware power and management:** power supply connector, Wake on LAN/Ring, fan connectors
- **Memory slots:** RIMM, DIMM (newer); SIMM, DIP (older)
- **Expansion slots:** ISA, PCI, AGP (most common)

Front Panel Connectors

- Hard disk LED (power/activity)
- Power on/off button, reset button
- +5V DC power, grounding
- System speaker connector
- Optional IR/IrDA serial port connector

Character Data (ASCII)

- **ASCII** = American Standard Code for Information Interchange
- Defines standard character set: letters, numbers, special/command/graphics characters

Character	Decimal	Binary
Null	0	00000000
Space	32	00100000
A	65	01000001
Z	90	01011010
a	97	01100001
z	122	01111010

Serial vs Parallel Data

- **Parallel data:** all bits of a character sent simultaneously over multiple wires
- **Serial data:** bits sent one at a time over a single wire
- More than 8 bits actually sent per character (extra bits used for data integrity, block ID, sync)

Transmission Modes

- **Simplex:** one direction only (e.g., speaker wire)
- **Half-duplex:** two directions, but one at a time (e.g., CB radio)
- **Full-duplex:** both directions simultaneously (e.g., telephone)

Serial Ports and Connectors

- Also called **COM port** or **RS-232 port**
- COM1, COM2, COM3... are logical names
- RS-232 = "Reference Standard 232" (IEEE wiring pattern standard)
- Connectors: **DB-9** (9-pin) or **DB-25** (25-pin), male on PC
- DB-25 also used for parallel ports and external modems

Pinouts (DB-25 vs DB-9)

DB-25 Pin	DB-9 Pin	Use
1	–	Ground
2	3	Transmit
3	2	Receive
4	7	RTS
5	8	CTS
6	6	DSR
7	5	Signal Ground
8	1	Carrier Detect
20	4	DTR
22	9	Ring Indicator

Asynchronous Communications

- Used by PC to connect printer, modem, fax, etc.
- No shared clock signal; devices operate independently
- Data blocks separated by idle periods
- **Data block format:** Start bit (0) + 8 data bits + parity bit (optional) + Stop bit (1) = 11 bits total
- **Parity:** forces count of 1-bits to even or odd; used for error checking
- Example: Letter "A" = 01000001, even parity bit = 0, stop bit = 1 ->
Transmitted block: 00100000101

The UART

- **UART** = Universal Asynchronous Receiver/Transmitter (pronounced "you-art")
- Controls serial ports; found on adapter card or motherboard
- Functions:
 - Controls connector pins/signals
 - Establishes communication protocol
 - Converts parallel (PC bus) to serial (transmission)
 - Converts serial (received) to parallel (PC bus)
 - Uses **FIFO** (First-In-First-Out) buffer
- **UART overrun**: buffer can't process incoming bits fast enough, causing data loss

UART Chip Table

Chip	Buffer Size (bytes)	Max Speed (bps)
8250	1	19,200
16450	1	38,400
16550	16	115,200
16650	32	430,800
16750	64	921,600
16850	128	1.5 Kbps
16950	128	1.5 Kbps

- Most modern PCs use **16550** UART (115.2 Kbps)
- UARTs above 16550 used in high-speed/multiport cards (ISDN, DSL modems)

Synchronous Communications

- Coordinated to a common clock (fixed interval between blocks)
- Sender's clock synced to receiver's; clock signal sent along with data
- Requires acknowledgment before next block sent

RS-232 Communications

- Official standard: EIA/TIA-232-E
- **DTE** (Data Terminal Equipment) = the PC
- **DCE** (Data Communications Equipment) = modem
- Handshake sequence:
- DTE sends signal on **DTR** (Data Terminal Ready)
- DCE responds via **DSR** (Data Set Ready)
- DTE sends **RTS** (Request to Send)
- DCE responds **CTS** (Clear to Send)
- Data flows via Transmit/Receive lines
- Flow control: RTS/CTS toggled off/on to pause/resume transmission

Cabling the Connection

- **Serial/modem/straight-through cable:** pins connected one-to-one (PC to modem)
- **Null modem cable:** cross-connected pins; used for DTE-to-DTE (PC-to-PC) connections

Configuring a Serial Port

- Default: COM1 (usually DB-9), COM2 (usually DB-25)
- COM1 and COM3 share an IRQ; COM2 and COM4 share an IRQ
- Multiport cards share a single IRQ with onboard processor managing traffic

Serial Port Resource Assignments

Logical Name	IRQ	I/O Address
COM1	IRQ 4	3F8h
COM2	IRQ 3	2F8h
COM3	IRQ 4	3E8h
COM4	IRQ 3	2E8h

Troubleshooting a Serial Port

- Common cause: system resource (IRQ) conflicts
- Checklist:
- Inspect for bent pins
- Ensure correct cable type (straight-through vs null modem)
- Check Device Manager for IRQ conflicts
- Cable should not exceed 50 feet

Parallel Ports

- Faster than serial (all bits of character sent at once)
- PC side: female DB-25; cable: male DB-25
- Originally for printers; later used for CD-ROMs, tape drives, Zip drives, file transfer

Parallel Port Standards (IEEE 1284)

- **SPP** (Standard Parallel Port): simplex, one direction only (PC to printer); supports legacy printers
- **EPP** (Enhanced Parallel Port): half-duplex, bidirectional but one direction at a time
- **ECP** (Enhanced Capabilities Port): full bidirectional, simultaneous communication; needs special IEEE 1284 cable

Parallel Port Resource Assignments

Port	IRQ	I/O Address	DMA Channel
LPT1	IRQ 7	378h	DMA 3 (ECP)
LPT2	IRQ 5	278h	n/a

Troubleshooting Parallel Ports

- Most problems lie in the device, not the port
- Check: bent pins/blocked holes, correct cable type (SPP/EPP/ECP), device itself
- Resource conflicts possible with new ECP hardware additions

The USB Interface

- **USB** = Universal Serial Bus (serial interface)
- Speeds: 12 Mbps (fast devices), 1.5 Mbps (slow devices); **USB 2.0** up to 480 Mbps
- Features:
- Plug-and-Play
- Standard connectors/cables (keyboards, mice, drives, printers, scanners, cameras, hubs)
- Automatic configuration
- Hot swapping (connect/disconnect while powered on)
- Supports up to 127 devices on one channel

USB Connectors

- **Type A**: host/hub side (permanently attached cables)
- **Type B**: peripheral side (detachable cable, squarish connector)

Speed Comparison (Mbps)

Interface	Speed
Ultra 3 SCSI	1,280
FireWire	800
Wide Ultra2 SCSI	640
USB 2.0	480
Ultra SCSI	160
Fast SCSI	80
SCSI	40
USB (high rate)	12
USB (low rate)	1.5
Serial	0.23

- Each USB port supplies 0.5 amps of power (enough for mouse/keyboard)

USB Interface Components

1. **USB host** – the PC (OS, chipset, BIOS support USB)

2. **USB hub** – allows tiered connections (hub-to-hub-to-device)
3. **USB devices** – peripherals connected directly or via hubs

How USB Works

1. Device plugged in; host/hub detects voltage change
2. **Enumeration**: host asks device to identify itself (type, manufacturer, function, bandwidth needed)
3. Device given unique address ID
4. Device driver loaded (prompt for disk/CD if unavailable)
5. Resource conflicts resolved automatically by host (no manual IRQ/DMA/I/O config needed)
6. On unplug: host detects voltage change, retires address ID, unloads driver
7. More info: www.usb.org

The FireWire Interface (IEEE 1394)

- High-speed serial interface bus
- Speeds: 100–400 Mbps (approx. 12–50 MBps)
- Newer versions (via 1394 Trade Association, www.1394ta.org) offer 800 Mbps to 1.6 Gbps